a. A message of Unsecured Session Type SHALL use the current *Global Unencrypted Message Counter*.

b. A message of Secure Unicast Session Type SHALL use the current *Secure Session Message Counter* for the session associated with the Session ID.

c. A message of Group Session Type SHALL use:

i. The *Global Group Encrypted Data Message Counter* if the Security Flags C Flag = `0`.

ii. The *Global Group Encrypted Control Message Counter* if the Security Flags C Flag = `1`.

2. The outgoing message counter from step 1 SHALL be incremented by 1.

3. Store the incremented outgoing message counter in the *OutgoingMessageCounter* element associated with the Session Context for the message.

a. If the message counter wraps around from 0xFFFF_FFFF to 0x0000_0000 and the message is of Secure Unicast Session Type:

i. The Encryption Key SHALL be expired in the Session Context. The session will need to be renegotiated to resume communication after transmission of this final message.

4.6.7. Counter Processing of Incoming Messages

1. Determine the Message Reception State for the sending peer and get the current `max_message_counter`.

a. Given a decrypted message of Unicast Session Type:

i. Get the session-specific Message Reception State from the Secure Unicast Session Context.

b. Given a decrypted message of Group Session Type:

i. Extract the Source Node ID from the Message Header.

A. If there is no Source Node ID for the message, drop the message.

ii. Get the Message Reception State for the Source Node ID of the given message:

A. If the Security Flags C Flag = 0, get the Data Message Reception State for the peer node.

B. If the Security Flags C Flag = 1, get the Control Message Reception State for the peer node.

iii. If there is no Message Reception State for the groupcast message, initiate Section 4.18.4, “Unsynchronized Message Processing”.

c. Given an unencrypted message:

i. Get the Message Reception State associated with the Unsecured Session Context.

ii. If there is no Message Reception State for the unencrypted message, create it with the information from the given message.

2. If the Message Counter is outside the valid message counter window, the message SHALL be marked as a duplicate. Note that while messages may be outside of the window for reasons other than being a duplicate, and we always mark them as such.

